

Slide 1

Toegepaste Informatica

2024



UCLL
HOGESCHOOL

- Why do games have so many bugs at launch?
- One puzzle piece :
 pygame -> open(tex).get_alpha()
 libglew.dll : [73]windows cannot access the file

Week 6: Textures



Quick detour

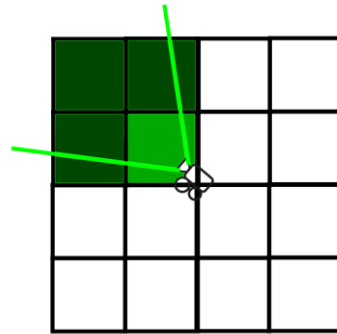
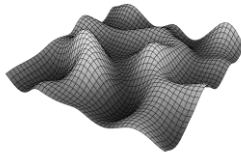
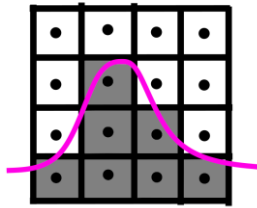
- Why do games have so many bugs at launch?
- One puzzle piece :
 pygame -> open(tex).get_alpha()
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Project

- Description will be feature complete Fri
- After that only clarifications and images
- How is it going?

Last class

- History of procedural generations (procgen)
- Seeing functions as geometry
- Generating geometry
- Infinite terrain



Course schedule

Highly likely to change ☹



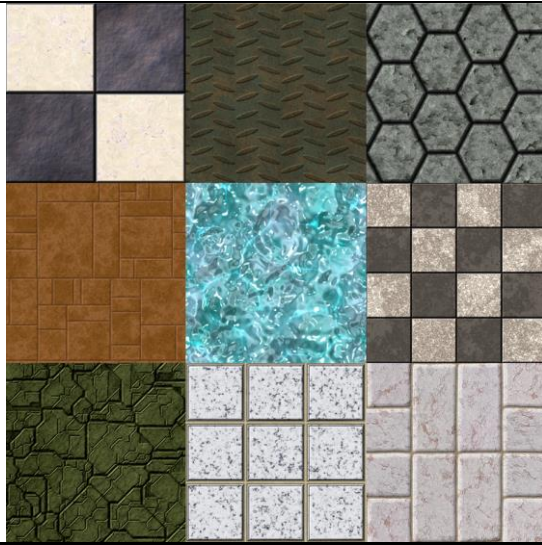
Week 1	Introduction
Week 2	3D reasoning
Week 3	Geometry
Week 4	Pipeline & Materials
Week 5	Proceduralism
Week 6	Textures & Normal maps
Week 7	Advanced pipeline
Week 8	Post Processing
Week 9	Aliasing / Presentations
Week 10 :	Project
Week 11 :	Revision

This class

- → Texture basic ←
- Texture types
- Border's, mips, and sampling
- Normal maps
- Creating normal maps

Textures

Can someone tell me what they are?



Textures

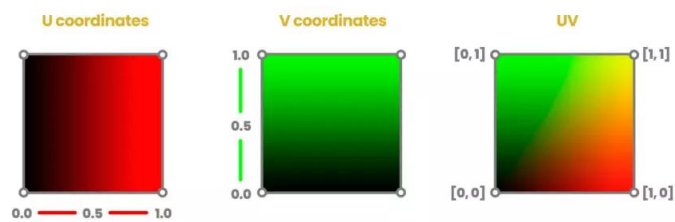
- Commonly : “a 2d grid of colors”
- 3d graphics : “discrete spatially varying information”

Textures : 2D grid of colors



Spatially varying

- Different areas in the texture have different values
- We use UV coordinates.
- Vector2D, 0.0 -> 1.0 (typically)
- They are part of the vertex

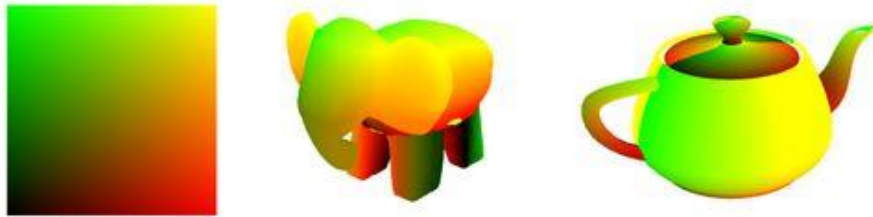


Ok, so the whole gist of it is, is that a texture (image) is a 2D grid of color values
To access these color values, we need an index. But we don't use integer indices
In the context of textures, we use floats, and we call them coordinate. The coordinates go from 0 to 1 in each dimension
An example would be the coordinate (0,0). This would fetch the bottom left color value

Do understand the images below. They're quite simple: they're visualizing the x coordinate in red, and the y coordinate in green over a square. In 3D graphics, we're often visualizing a ton of data in space, and it can be very useful to use color for that.

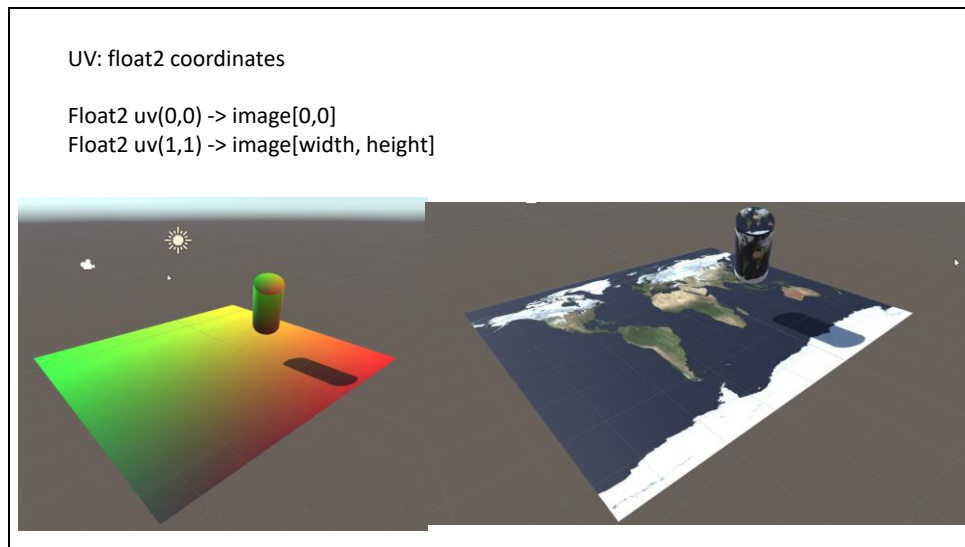
2D coordinate system on a 3D model

- Most common way to put textures on 3D models
- Referred to als "uv mapping"



When people make models, they need to put uv coordinates in the vertices, so we can figure out where parts of the texture goes.

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The plane has obvious texture coordinates, and when we put the world map on the plane using these coordinates, it looks like we expect

On the cylinder, we have the same world map, but it's wrapped around the geometry.

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I guess we've always known how this works, just the other way around

A model is 3 dimensional, but a texture is 2 dimensional

Our vertices, with 3 dimensional positions, have another set of coordinates, in 2D, called uv coordinates. We need those so we know where our 2D texture goes in 3D space.

In code

```
// in Properties
_MainTex ("Albedo (RGB)", 2D) = "white" {}

// in SubShader

sampler2D _MainTex;

struct Input
{
    float2 uv_MainTex;
};

// in fragment shader
fixed4 color = tex2D (_MainTex, IN.uv_MainTex)
```

In code, this is dead simple

We have a texture, which is the 2D array of color values

We also have a sampler. A sampler we'll see later, but we need a sampler to get use our coordinates to access the texture

Why not put this information in a vertex?

- Good question!
- Vertex density vary with geometric complexity (wheels have more vertices than doors)
- Textures have a more homogenous resolution



This is a bad example. But imagine we would have a 3D model of the cybertruck. We would only need very few triangles to store the geometry (position of the triangles)

The graffiti, we would need a big texture to capture all the detail and the curved lines.

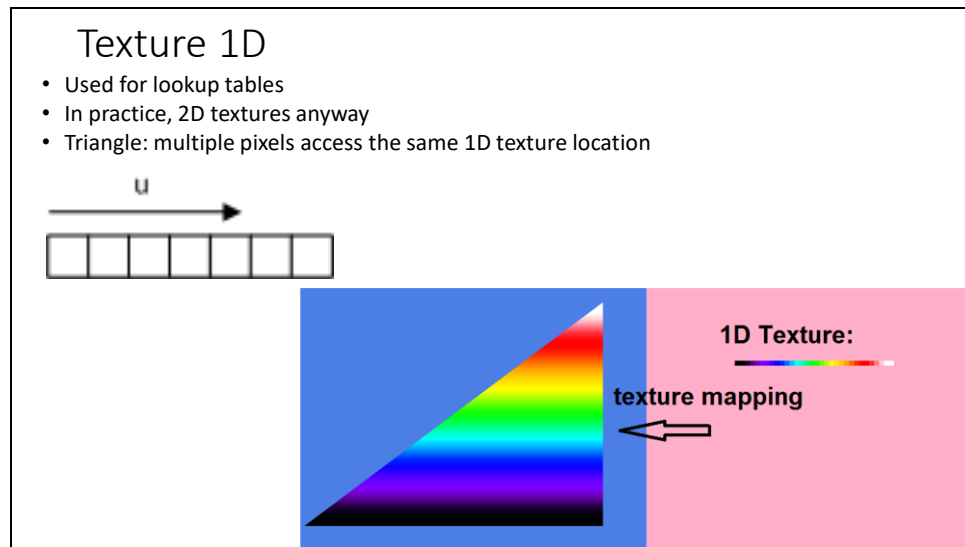
Any questions?

This class

- Texture basic
- → Texture types ←
- Border's, mips, and sampling
- Normal maps (for normal students ©Margot Schepers)
- Creating normal maps

We just saw 2D textures as an example

- Any guess what the others types are?



1D textures are almost exactly the same as 2D textures, but the coordinate is just a simple float

This is often used as a lookup table : for example, what is the color of the sun over a 24 hour period.

The texture above could be made to make an incorrect rainbow: a rainbow changes color top to bottom, but not left to right

